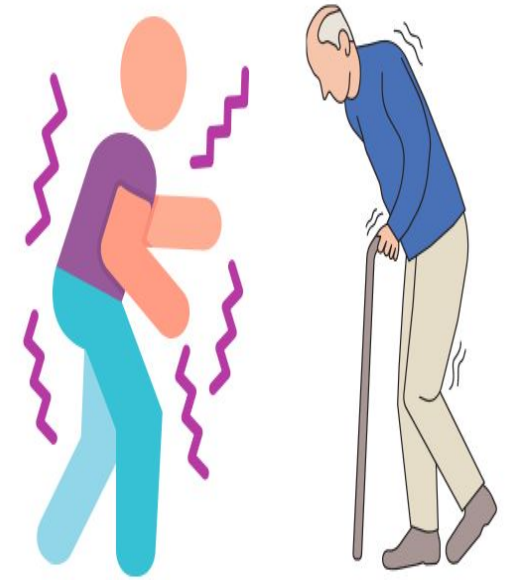


Classify movement disorders, explain physiological basis and relate disorders to neural dysfunction.



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Movement disorders are a group of neurological conditions that cause abnormal movements. They could be increased movement (like spasms, jerking or shaking) and/or decreased or slow movement. They can affect actions you choose to make (voluntary) or cause uncontrolled (involuntary) movements.

Movement disorders are clinical syndromes categorized broadly into hypokinetic (reduced/slow movement) and hyperkinetic (excess/involuntary movement) states, arising fundamentally from structural or functional disruptions within the basal ganglia and associated extrapyramidal motor circuits

Every body movement you make, from lifting your leg to moving your jaw and tongue to talk, involves complex communication between your:

- Central nervous system (your brain and spinal cord)
- Motor nerves
- Skeletal muscles

Movement Disorders

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graph TD; A[Movement Disorders] --> B[Hypokinetic]; A --> C[Hyperkinetic]; B --> B1[• Parkinson disease]; B --> B2[• Parkinson plus]; B --> B3[• Secondary parkinsonism - Drug-induced]; C --> C1[• Tremor-Essential Tremor]; C --> C2[• Chorea-Huntington Disease]; C --> C3[• Dystonia]; C --> C4[• Tics-Tourette]; C --> C5[• Wilson Disease]; C --> C6[• Myoclonus];
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Hypokinetic

- **Parkinson disease**
- Parkinson plus
- Secondary parkinsonism
 - Drug-induced

Hyperkinetic

- Tremor-Essential Tremor
- Chorea-Huntington Disease
- Dystonia
- Tics-Tourette
- Wilson Disease
- Myoclonus

Classification of Movement Disorders

- **Hyperkinetic (excess movement):**

- **Tremor:** Rhythmic, shaking movements in parts like hands.
- **Chorea:** Brief, rapid, jerky, unpredictable movements.
- **Dystonia:** Sustained, twisting muscle contractions and abnormal postures.
- **Myoclonus:** Sudden, shock-like muscle jerks.
- **Tics:** Sudden, repetitive, stereotyped movements or sounds.

- **Hypokinetic (reduced movement):**

- **Parkinsonism:** Characterized by bradykinesia (slowness), rigidity, and resting tremor.
- **Akinesia/ Akinetic crisis:** Complete or near-complete loss of voluntary movement.

Physiological Basis of Movement Control

Physiological Basis & Neural Dysfunction

Movement control relies on a delicate balance between the direct pathway (which promotes movement via cortical excitation) and the indirect pathway (which inhibits movement) through the basal ganglia loop (striatum, globus pallidus, subthalamic nucleus, and substantia nigra).

The Basal Ganglia Loop:

A group of deep brain structures (striatum, globus pallidus, subthalamic nucleus, substantia nigra) that work with the cortex and thalamus.

- **Direct and Indirect Pathways:**

- **Direct pathway:** Facilitates and promotes desired voluntary movements.
- **Indirect pathway:** Inhibits and suppresses unwanted, competing movements.

- **Neurotransmitter Balance:**

- Dopamine acts as a gas pedal for the direct pathway and a brake for the indirect pathway, while acetylcholine provides a balancing opposing tone.

Relation to Neural Dysfunction

Hypokinetic Disorders (e.g., Parkinson's Disease):

- Caused by the loss of dopamine-producing cells in the *substantia nigra pars compacta*.
- This shifts balance to over activity in the indirect pathway and excessive inhibition of the thalamus.
- Result: The brain struggles to start and execute movements (bradykinesia, rigidity).

Hyperkinetic Disorders (e.g., Huntington's Disease, hemiballismus):

- Hyperkinetic movement disorders comprise a range of diseases characterized by unwanted and uncontrollable, or poorly controllable, involuntary movements.
- Caused by damage or loss of neurons in specific parts of the striatum or the subthalamic nucleus.
- This disrupts the normal inhibitory control of the indirect pathway or subthalamic circuit.
- Result: Unwanted motor signals leak through, causing abnormal extra movements like chorea or wild flailing.

Parkinson's disease (PD)

- Parkinson's disease (PD) is a chronic, progressive neurodegenerative disorder characterized by slowness in the initiation and execution of movement (bradykinesia), increased muscle tone (rigidity), tremor at rest, and gait changes.

Typical appearance of Parkinson's disease



What are the types of abnormal movements?

There are two main types of abnormal movements:

- Hyperkinetic movement.
- Hypokinetic movement.

A movement disorder can have both of these or just one

Hyperkinetic movement

Hyperkinetic movement disorders involve increased movement. “Hyper” means “over” or “beyond,” and “kinetic” means “motion.” It can affect voluntary movement (actions you choose to take) or cause involuntary movement (actions that are out of your control).

Types of hyperkinetic movement include:

- **Akathisia**
- **Ataxia**
- **Chorea, athetosis and ballism**
- **Athetosis**
- **Dystonia**
- **Myoclonus**
- **Spasticity**
- **Stereotypies (stereotypic movements)**
- **Tics**
- **Tremor**

Akathisia: Akathisia is a movement disorder that causes an intense, uncontrollable inner restlessness and a strong urge to move

Akathisia makes it difficult to sit or remain still due to inner restlessness.

Akathisia is associated with the use of certain types of medications, especially antipsychotic (neuroleptic) medications.

AKATHISIA

DEFINITION

Akathisia is an extrapyramidal adverse effect characterized by a subjectively distressing sense of inner restlessness accompanied by an urge or compulsion to move.

Patients often describe an inability to remain still.



CLINICAL FEATURES



SUBJECTIVE SYMPTOMS



Inner restlessness



Feeling unable to sit still



Strong urge to keep moving



Marked distress or discomfort



Akathisia can be mistaken for psychotic agitation or anxiety, leading to inappropriate treatment.



OBJECTIVE SIGNS



Foot stamping while seated



Repeated crossing and uncrossing of the legs



Rocking from one foot to the other



Constant pacing



Frequent shifting of posture



Inability to remain seated for prolonged periods

Ataxia

Ataxia is a neurological sign and medical term meaning a lack of voluntary coordination of muscle movements. It affects balance, walking, speech, and fine motor skills, often making a person's movements look clumsy, unsteady, or similar to being intoxicated

Types of ataxia

Ataxia, or uncoordinated movements, is categorized in different ways:



CEREBELLAR
There's a problem in the cerebellum, a part of your brain.



SENSORY
There's a disruption in your "self-positioning" sense (proprioception).



VESTIBULAR
There's a problem with your inner ears, which are part of your sense of balance.



ACQUIRED
You develop it at some point in your life.



INHERITED
These are genetic conditions, meaning you inherit them from one or both of your biological parents.



SPORADIC
These happen because of DNA changes that aren't inherited.

These types can overlap. For example, ataxia from a stroke in your cerebellum would be acquired cerebellar ataxia.

ATAXIA



Chorea, athetosis, and ballism

Hyperkinetic movement disorders caused by dysfunction of the basal ganglia. They involve continuous, involuntary, and random muscle contractions that differ primarily in speed, rhythm, and the specific muscle groups they affect.

Chorea

The most common symptoms are:



Grip changes.



Involuntary muscle movements.



Tongue movements.

Children may have the following symptoms in addition to involuntary muscle movements:



Headaches.



Slurred speech.



Seizures.

Athetosis: is a symptom defined by slow, continuous, involuntary, and writhing movements—often described as worm-like or snake-like—that mostly affect the hands, fingers, feet, and toes. It makes holding a stable posture or performing precise tasks very difficult.

ATHETOSIS

Involuntary, Slow & Writhing Movements



WHAT IS ATHETOSIS?

Athetosis involves involuntary and continuous twisting, writhing movements of the limbs and extremities, often associated with damage to the Basal Ganglia in the brain.



DOPAMINE ↓ ↑ ACETYLCHOLINE

Dopamine Acetylcholine

CAUSES

- ✓ Cerebral Palsy
- ✓ Perinatal brain injury
- ✓ Stroke
- ✓ Hypoxia
- ✓ Kernicterus
- ✓ Genetic conditions (e.g. Wilson's disease)



MOVEMENT CHARACTERISTICS

- ✓ Uncontrolled, purposeless, writhing movements
- ✓ Face, hands, arms, legs, and sometimes trunk involved
- ✓ Movements slow, continuous, irregular
- ✓ Overflow movements may occur



OBJECTIVE FINDINGS

- ✓ Functional limitations
- ✓ Speech difficulties
- ✓ Muscle weakness
- ✓ Difficulty with fine motor tasks



MANAGEMENT

- ✓ Physical therapy for strengthening
- ✓ Occupational therapy for ADLs
- ✓ Speech therapy
- ✓ Assistive devices if necessary



Dystonia is a neurological movement disorder that causes involuntary muscle contractions. These contractions lead to twisting, repetitive movements, or abnormal, fixed postures that can range from mild to painful.



Findings of Dystonia Disease are;

Oromandibular Dystonia or Meige Syndrome

Involuntary opening, closing, or deviation of the jaw

Blepharospasm

Involuntary contractions of the muscles around the eyes

Spasmodic Dysphonia or laryngeal Dystonia

Affects the vocal cords to have strangled, hoarse quality or a breathy, whispering voice

Cervical Dystonia or Spasmodic Torticollis

Affects the neck muscles leading to abnormal movements of the head and neck

Advantages

- ✓ Adjustable
- ✓ Reversible
- ✓ Rechargeable

Writers cramp, Musician's Dystonia

Involuntary movements, cramping and spasming of the hands or arms

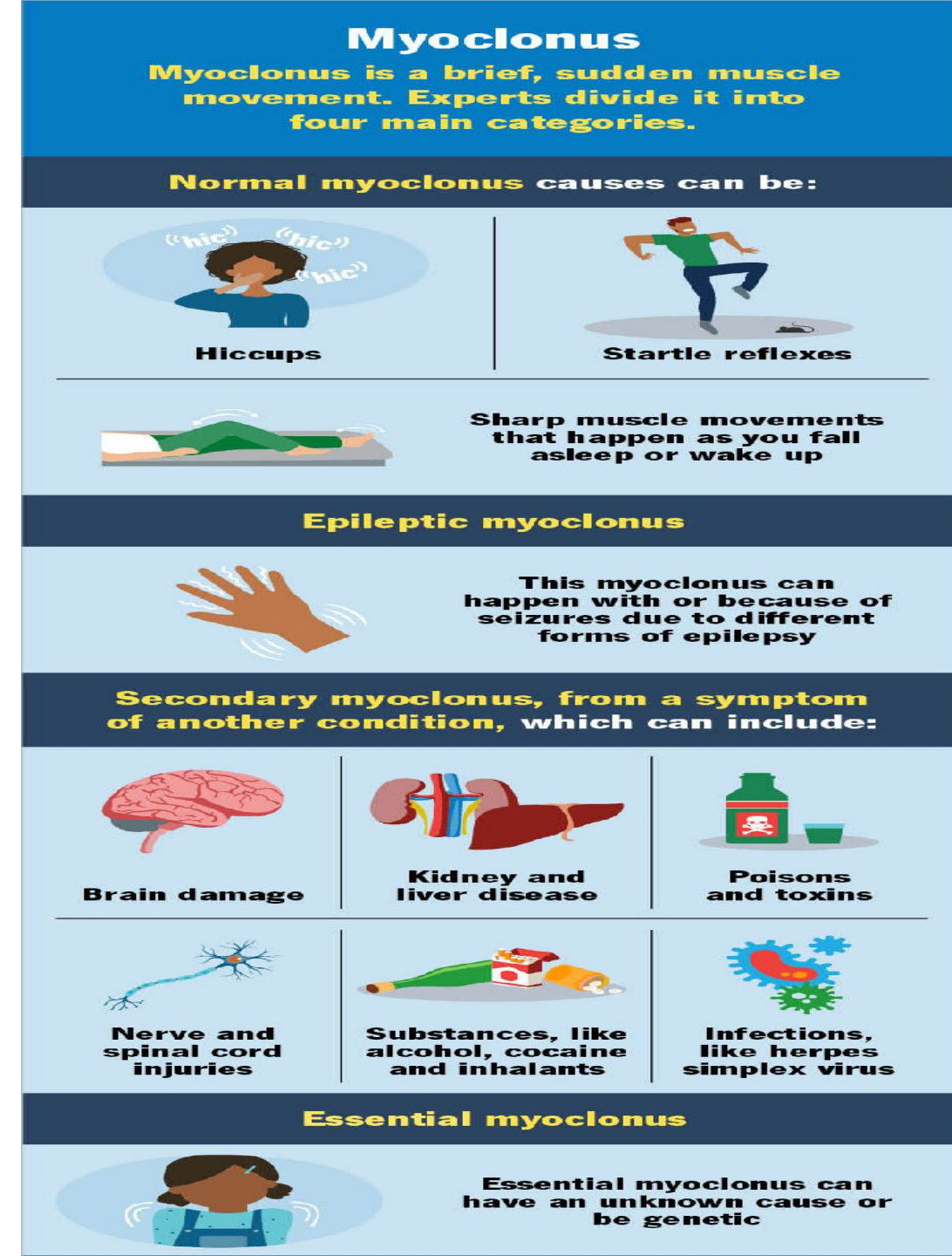
Limb Dystonia

Involuntary movements, cramping and spasming of the legs or feet

Deep Brain Stimulation Surgery is an effective treatment method for Dystonia

What is myoclonus?

- Myoclonus is a type of uncontrollable movement that includes sudden, brief involuntary twitching, jerking, or spasm of a single muscle or a group of muscles.
- A person experiencing myoclonus cannot control it. Myoclonus is the word for the movements themselves. It is not a disease itself, but may be a sign of another neurological condition.
- Causes of myoclonic twitches or jerks are:
- Sudden muscle contractions (tightening), called **positive myoclonus**
- Sudden muscle relaxation, called **negative myoclonus**



Stereotypic movement disorder (SMD) involves repetitive and purposeless movements, like hand flapping.

It has no known underlying cause, and the movements interfere with your child's daily activities.

The movements may also cause self-injury.

Treatment for SMD isn't necessary, but habit reversal training can help



Tics

Tics are sudden, quick movements or sounds that you do repeatedly and involuntarily.

If a person have tics, you can't stop your body from doing these things. They can occur in any part of your body, including your face, shoulders, hands and legs.

For example, if a person have a motortic, he or she might blink your eyes repeatedly or shrug your shoulders over and over again.

If a person have a vocal tic, he or she may make a grunting sound unwillingly or feel the need to constantly clear your throat. The tics you do often change over time.

Tics and tic disorders

Tics are sudden, quick movements or sounds that you do repeatedly and involuntarily.

Simple motor tics can include:



Facial grimacing.



Eye blinking and other eye movements.



Shoulder shrugging.

Simple vocal tics can include:



Barking.



Repetitive throat clearing.



Grunting.

Complex motor tics can include:



Jumping.



Repeatedly sniffing an object.



Twisting.

Complex vocal tics can include:



Repeating your own words or phrases.



Repeating other people's words or phrases.



Using obscene or swear words.

Hypokinetic movement

- Hypokinetic movement disorders involve decreased or slow movement. “Hypo” means “below” or “beneath.” It generally affects voluntary movement.
- **Parkinsonism** is the main type of hypokinetic movement. It’s an umbrella term that refers to brain conditions that cause slowed movements, rigidity (stiffness), tremor or balance trouble.
- Neurodegenerative parkinsonism most commonly happens with Parkinson’s disease, but it can also be a feature of the following conditions:
 - Multiple system atrophy.
 - **Corticobasal degeneration.**
 - Progressive supranuclear palsy.
 - **Lewy body dementia**
- **Parkinsonism** can also develop due to repeated head injuries, toxic substances, use of psychiatric medications and a lack of blood flow to certain areas of your brain.
- **Bradykinesia** is another hypokinetic movement. It involves slowness of movement and speed or progressive hesitations or halts as you continue movements. It’s one of the main signs of Parkinson’s disease.

Tremor:-

Tremor is a rhythmic movement of a body part that's out of your control (involuntary).

The movements look like trembling or shakiness and can range from mild to severe. They may be constant or only happen sometimes.

Tremor is a common symptom of movement disorders.

You can have tremor in one part of your body or multiple parts. It most commonly affects your hands. But it can also occur in your:

- Arms.
- Legs.
- Head, neck or face.
- Vocal cords.



What are the types of movement disorders?

There are several movement disorders. Some include:

- **Essential tremor:** Essential tremor causes parts of your body to shake uncontrollably. It usually affects your hands and arms but can also affect your head, voice and other body parts.
- **Huntington's disease:** Huntington's disease is a genetic condition that affects your brain, causing unsteady and uncontrollable movements (chorea) in your hands, feet and face. Symptoms get worse over time.
- **Multiple system atrophy (MSA)** MSA is a rare condition that causes certain brain areas to deteriorate. It may cause ataxia and parkinsonism.
- **Parkinson's disease:** Parkinson's disease is a brain disorder that causes unintended or uncontrollable movements and difficulty with balance and coordination. It also causes cognitive (mental) decline.
- **Periodic limb movement disorder (PLMD):** PLMD involves repetitive limb movements that occur during sleep and may cause sleep disruption. The limb movements usually involve your lower extremities, consisting of extension of your big toe and flexion of your ankle, knee and hip.
- **Progressive supranuclear palsy (PSP):** PSP is a rare neurodegenerative disorder that damages certain areas of your brain. It affects how you walk, think, swallow and move your eyes.
- **Restless leg syndrome (RLS):** RLS is a sleep disorder that causes an intense, often irresistible urge to move your legs (and even your arms or body). It occurs along with other sensations in your limbs like pulling, creeping, tugging, throbbing, itching, aching, burning or crawling.
- **Rett syndrome:** Rett syndrome is a rare genetic condition that affects speech, purposeful hand use and coordination.
- **Tardive dyskinesia:** Tardive dyskinesia can result from taking antipsychotic (neuroleptic) medications for many years. It can cause involuntary facial tics. It can also cause uncontrollable movements like lip-smacking.
- **Tourette syndrome:** Tourette syndrome is a neurological disorder that affects your brain and nerves. It causes you to make sudden movements or sounds (tics) and is accompanied by anxiety, ADHD and often obsessive-compulsive disorder.
- **Wilson disease:** Wilson disease is a rare genetic condition that happens when your body accumulates too much copper, especially in your liver and brain. It can cause stiff muscles, tremors and uncontrolled movements.

Common Movement Disorder Symptoms



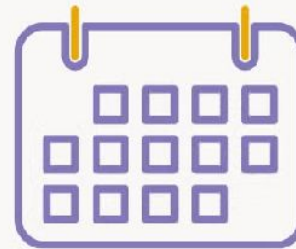
Involuntary movements

Tremors, jerky motions, or sustained muscle contractions



Impaired daily function

Difficulty with simple tasks, such as walking and writing



Progression of symptoms

Symptoms tend to worsen over time



Interference with cognitive function

Decline in memory and decision-making

Symptoms and Causes

What are the symptoms of movement disorders?

The symptoms of movement disorders vary widely. All movement disorders cause abnormal movements. Some movement disorders have other symptoms, such as thinking and mood changes. The symptoms can range in severity from barely noticeable to disruptive.

In general, signs and symptoms of movement disorders include:

- Episodes of uncontrolled movement, like twitches, spasms, tremors, jerks, twisting and shaking.
- Problems with coordination and balance.
- Trouble with certain movement tasks, such as writing, swallowing or speaking.
- Difficulty walking or changes to your gait.
- Stiffness or rigidity of your limbs and trunk.

Abnormal movements can affect one or more of several parts of your body, including your:

- Limbs: Hands and fingers. Feet and toes.
- Facial muscles.
- Head and neck.
- Trunk and posture.
- Voice.

What is the cause of movement disorders?

- In general, movement disorders develop from damage to or malfunction of certain parts of your brain that control movement, including your:
- **Primary motor cortex:** The primary motor cortex is a strip of brain tissue in your frontal lobe. It's responsible for initiating voluntary movements. Damage to your primary motor cortex can result in spasticity, myoclonus and issues with fine motor movements.
- **Basal ganglia:** Your basal ganglia help start and smooth out voluntary muscle movements, suppress involuntary movements and coordinate changes in your posture. Damage to your basal ganglia can result in chorea, athetosis, dystonia and parkinsonism.
- **Cerebellum:** Your cerebellum coordinates your movements, helps your limbs move smoothly and accurately, and helps maintain balance. Damage to this part of your brain results in loss of coordination.
- **Thalamus:** Your thalamus is an egg-shaped structure in the deeper part of your brain — one on either side. It's a relay station of all incoming motor (movement) and sensory information. Damage to your thalamus can cause tremor and motor impairments.

Other Causes

A wide variety of factors can cause movement disorders, including:

Genetics. Some types of movement disorders can be caused by an altered gene. The altered gene is passed down from a parent to a child. This is called an inherited condition. Huntington's disease and Wilson's disease are two movement disorders that can be inherited.

Medicines. Medicines such as anti-seizure and anti-psychotic medicines can lead to movement disorders.

Illegal drugs or large amounts of alcohol. Illegal drugs such as cocaine can cause movement disorders such as chorea. Drinking large amounts of alcohol can cause chorea or ataxia.

Not getting enough of certain vitamins. Having low levels of certain vitamins in the body, known as a vitamin deficiency, can cause movement disorders. A deficiency in vitamin B-1, vitamin B-12 or vitamin E can lead to ataxia.

Medical conditions. Thyroid conditions, multiple sclerosis, stroke, viral encephalitis and several others can cause movement disorders. Brain tumors also can lead to movement disorders.

Head injury. Head trauma from an injury can lead to movement disorders.

For most people with a movement disorders, there is no known cause. When healthcare professionals haven't found the exact cause, it's called idiopathic.

Diagnosis and Tests

How are movement disorders diagnosed?

As movement disorders are often complex and mimic other conditions.

Doctor likely will perform multiple tests to make a diagnosis.

They'll first start with a detailed history, **physical exam** and a **neurological exam**

Based on your symptoms, they may order any of these tests:

- **Blood tests** to help diagnose certain kinds of movement disorders or rule out other causes.
- **Electromyography (EMG)** to assess the health of your muscles and the nerves that control them.
- **Electroencephalogram (EEG)** to check the electrical activity of your brain.
- **Lumbar puncture** to analyze your cerebrospinal fluid.
- Muscle **biopsy** to distinguish between nerve and muscle conditions.
- Nerve conduction study to measure the flow of electrical current through a nerve before it reaches a muscle.
- Providers also often use imaging tests to help diagnose movement disorders. They may look at your brain, spinal cord or nerves. These tests may include:
 - **Computed tomography (CT)**
 - **Magnetic resonance imaging (MRI)**

Management and Treatment

How are movement disorders treated?

- The treatment for movement disorders varies based on the type. Most movement disorders don't have a cure, so the goal of treatment is to manage symptoms. But some movement disorders, such as medication-induced parkinsonism, are often treatable.

Examples of treatments for movement disorders include:

- **Medication:** Several medications can help the symptoms of movement disorders. For example, muscle relaxants can help with spasticity. Dopaminergic medications may help with Parkinson's disease and restless leg syndrome. Antianxiety medications may help with dystonia. There are also specific medications for specific conditions.
- **Physical therapy** Physical therapy helps improve how your body performs physical movements. Physical therapists help you manage symptoms like pain, stiffness and discomfort that make it hard to move.
- **Occupational therapy** Occupational therapy helps improve your ability to perform daily tasks. An occupational therapist helps you learn how to safely stand, sit, move or use different tools to participate in your activities.
- **Mobility aids** Mobility aids, like canes, walkers and wheelchairs, can help you move more safely and increase your independence.
- **Speech therapy** helps improve your speech, language skills and swallowing ability.

Management and Treatment (Continued)

- **Psychotherapy** (talk therapy) is a term for a variety of treatment techniques that aim to help you identify and change unhealthy emotions, thoughts and behaviors. Movement disorders often lead to mental health conditions, like **depression** and **anxiety**. Psychotherapy can help.
- **Botulinum toxin (Botox®) injections** These can help relax your muscles if you have dystonia or spasticity.
- **Deep brain stimulation** This is a brain surgery for people with advanced Parkinson's disease, dystonia and other tremors. It may reduce involuntary movements.
- **Clinical trials**: Clinical trials are a type of research that studies new tests and treatments for specific conditions. There may be a clinical trial you can be a part of.

How do I know if I have a movement disorder?

We all experience uncontrollable movements at times, such as random muscle twitches, hypnic jerks or hiccups.

However, unusual or persistent symptoms may be signs of a movement disorder.

If a person notice a consistent change in your or your child's movements, it's important to see a healthcare provider to receive a diagnosis and treatment.

8 MOVEMENT DISORDER LOCALIZATION



LOCALIZE THE LESION, FIND THE CAUSE!

1 BASAL GANGLIA  MOVEMENT DISORDERS • Parkinsonism • Chorea (Huntington's) • Dystonia CLUES Bradykinesia, rigidity, resting tremor, chorea, dystonic postures COMMON CAUSES Parkinson's disease, Huntington's disease, drugs, stroke	2 THALAMUS  MOVEMENT DISORDERS • Tremor • Hemiballismus CLUES Contralateral tremor, hemiballismus COMMON CAUSES Stroke, hemorrhage, multiple sclerosis	3 CEREBELLUM  MOVEMENT DISORDERS • Ataxia • Intention tremor • Dysmetria CLUES Gait ataxia, dysmetria, nystagmus, dysidiadochokinesia COMMON CAUSES Stroke, tumors, MS, alcohol, drugs	4 BRAINSTEM  MOVEMENT DISORDERS • Tremor • Dystonia • Parkinsonism (rare) CLUES Associated cranial nerve signs, long tract signs COMMON CAUSES Stroke, tumors, neurodegenerative diseases
5 SPINAL CORD  MOVEMENT DISORDERS • Spasticity • Clonus • Segmental myoclonus CLUES LMN signs below lesion, sensory level COMMON CAUSES Trauma, MS, tumor, transverse myelitis	6 CORTEX (MOTOR)  MOVEMENT DISORDERS • Focal motor seizures • Apraxia • Myoclonus CLUES Contralateral weakness, seizures, cortical signs COMMON CAUSES Stroke, tumor, encephalitis	7 PERIPHERAL NERVES  MOVEMENT DISORDERS • Fasciculations • Weakness • Tremor (rare) CLUES LMN signs: weakness, atrophy, fasciculations COMMON CAUSES Neuropathy, entrapment, motor neuron disease	8 NEUROMUSCULAR JUNCTION / MUSCLE  MOVEMENT DISORDERS • Fatigable weakness • Myasthenia gravis • Myopathies CLUES Ptosis, diplopia, fatigability, proximal weakness COMMON CAUSES Myasthenia gravis, muscular dystrophies, myositis